FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

HALF COURSE TEST –VIII

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
   Section – A (05 – 09) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.
3. **Section – C (01 – 08)** contains 8 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION - A

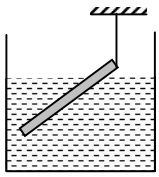
Straight Objective Type

*This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

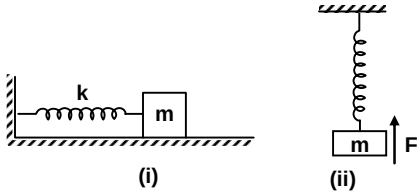
1. A block of mass m attached to a spring executes SHM with angular frequency ω_0. The spring is cut into 4 equal parts and another block with mass $\frac{m}{2}$ is attached to the two pieces in parallel combination. The angular frequency of oscillations of the mass will be
(A) $2\omega_0$ (B) $4\omega_0$
(C) $2\sqrt{2}\omega_0$ (D) $\omega_0/4$

2. An artificial satellite is first taken to a height equal to half the radius of Earth. Let E_1 be the energy required. It is then given the appropriate orbital speed such that it goes in a circular orbit at that height. Let E_2 be the energy required. The ratio $\frac{E_1}{E_2}$ is
(A) 4 : 1 (B) 3 : 1
(C) 1 : 1 (D) 1 : 2.

3. A thin homogeneous rod of length 2L floats partly immersed in water, being supported by a string fastened to one of its end, as shown. The specific gravity of the rod is 0.75. the length of the rod that extends out of water in equilibrium is
(A) L (B) L/2
(C) L/4 (D) 3L



4. A particle of mass m is executing SHM as shown in diagram (1) and (2). In diagram (2) a force F is applied on the particle in the upward direction F = mg. If their velocities are same at mean position and let A_1 and A_2 be their amplitudes and T_1 and T_2 be their time periods respectively then,
(A) $A_1 = A_2$, $T_1 = T_2$
(B) $A_1 = A_2$, $T_1 > T_2$
(C) $A_1 > A_2$, $T_1 = T_2$
(D) $A_1 > A_2$, $T_1 > T_2$



Space for rough work

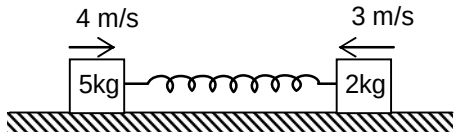
(Multiple Correct Choice Type)

This section contains ***5 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

5. A hollow sphere and a solid sphere of same radius and same material fall through a liquid from same height (neglect viscous force). Then, the correct option(s) from the following is/are:
   (A) The solid sphere reaches the ground earlier than the hollow sphere.
   (B) The hollow sphere reaches the ground earlier than the solid sphere.
   (C) Buoyant forces on both solid as well as hollow sphere are same.
   (D) Buoyant forces on the solid and hollow spheres are different.

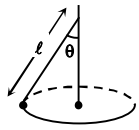
6. A spherical black body has radius r. If P is the power radiated by the black body and R is the rate of cooling of the black body, then
   (A) $P \propto \frac{1}{r}$ (B) $P \propto r^2$
   (C) $R \propto \frac{1}{r}$ (D) $R \propto r^2$

7. Two blocks connected with the spring of force constant 100 N/m are given velocities $4\hat{i}$ m/s and $-3\hat{i}$ m/s when the spring is in natural length as shown in figure



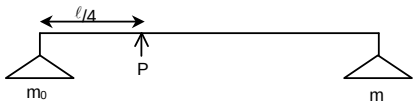
   (A) The velocity of 2kg block is maximum when 5 kg block is at instantaneous rest.
   (B) The maximum and minimum velocities of 2kg block is $7\hat{i}$ m/s and $-3\hat{i}$ m/s respectively.
   (C) The maximum and minimum velocities of 5kg block is 4 m/s and zero respectively.
   (D) None of these

8. A particle is rotating in a conical pendulum with help of a string of length ℓ as shown in the figure. The speed of the particle is constant and angle θ is also constant with time. It can be said that



   (A) Angular momentum of the particle about the point of suspension is not constant.
   (B) Only the direction of angular momentum of the particle about the point of suspension is constant.
   (C) Only the magnitude of angular momentum about the point of suspension is constant.
   (D) Net torque on the particle about the point of suspension is zero.

9. A light rod of length ℓ is pivoted at distance $\ell/4$ from the left end has two masses m_0 and m attached to its ends such that rod is in equilibrium. Find m in terms of m_0.



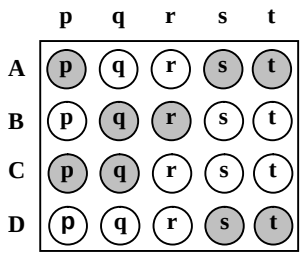
   (A) $\frac{m_0}{2}$ (B) $\frac{m_0}{3}$
   (C) Torque about P is zero (D) Torque about P is not zero

Space for rough work

SECTION - B

Matrix – Match Type

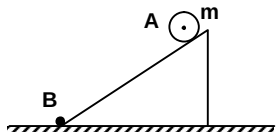
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

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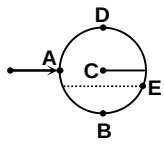
1. A sphere of mass m, radius R is left on a rough wedge (coefficient of friction μ) fixed with the ground. Now, match the following



Column A		Column B	
(A)	If $\mu = 0$	(p)	ΔP.E. > ΔK.E.
(B)	If $\mu \neq 0$; and before rolling starts	(q)	Friction is acting upward
(C)	If $\mu \neq 0$; and after rolling starts	(r)	Friction is acting downward
(D)	If $\mu \neq 0$; when the sphere is at bottom most point.	(s)	ΔP.E $\leq$ ΔK.E.
		(t)	Translational (ΔK.E.) > (Rotational) (ΔK.E.)

Space for rough work

2. A sphere of mass M, radius R kept on a horizontal frictionless surface. A small ball of mass m hits the sphere at a height h = R and rebounds elastically as shown in the figure. Consider the situation mentioned and match the following.



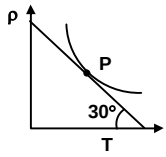
	Column A		Column B
(A)	Linear momentum for the sphere is conserved along tangent at	(p)	A
(B)	Angular momentum for the system is conserved about point.	(q)	B
(C)	Net torque on the sphere is zero about	(r)	C
(D)	Linear Impulse is acting about line joining points A and	(s)	D
		(t)	E

Space for rough work

SECTION – C
(Integer value correct Type)

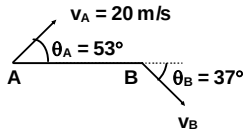
This section contains ***8 questions***. The answer to each question is ***a single digit integer*,** ranging from 0 to 9 (*both inclusive*).

1. The variation of density (ρ) with temperature (T) of an ideal gas of molecular mass (M) at constant pressure is plotted as shown in figure.



The pressure of the gas at state P is $\frac{RT^2}{\sqrt{KM}}$ Find the value of K.

2. The two ends A and B of a uniform of length ℓ = 1m and mass m are moving with velocities V_A and V_B as shown. The length AP, where P is point on the rod with velocity 12 m/s, is 8K cm. Find the value of K.



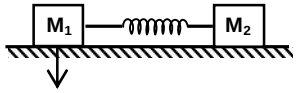
3. A particle moving on the sphere $X^2 + Y^2 + Z^2 = 1$ on the convex side. XY plane is vertical plane and 'g' acts along –ve Y axis. The speed of the particle is $2ms^{-1}$ always. The mass of the particle is 1kg. The normal reaction acting on the particle at point (0, 1m, 0) is _____.

4.



Frictionless surface

If M_1 & M_2 are free to oscillate together about their centre of Mass, then time period is T_1.



M_1 fixed

If M_1 is fixed, and only M_2 oscillates horizontally then time period is T_2

If $\frac{T_2}{T_1} = 3$ Then $\frac{M_2}{M_1}$ is _____

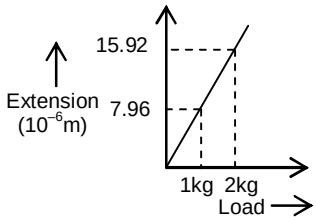
Space for rough work

5. The displacement of a string is given by $y(x_1,t) = .06\sin\left[\frac{2\pi X}{3}\right]\cos[120\pi t]$

Where X and Y are in m and t in sec. The length of the string is 1.5m and its mass is 3.0×10^{-2} Kg. This is a result of super position of two waves travelling in opposite directions.
The speed of any one of the component waves is 36X ms^{-1}. Then X = ?

6. A hollow sphere of volume V is floating on water surface half immersed. The minimum volume of water poured on surface so that the sphere now sinks into the water is V/K. Then K is:

7. In determination of Young's Modulus by Searle's Method, the extension Vs load curve is shown below. The radius of the wire is 1mm and length of the wire used is 2m. The Young's modulus is $X\times10^{11}N/m^2$. Then X = _____.



8. A particle starts falling from the top of a tower at t = 0 where g = 10ms^{-2}. Simultaneously, a car situated on ground some distance away starts moving towards the tower at a constant velocity of 5ms^{-1}. The radius of curvature of the trajectory of particle as seen by an observer in the Jeep at some time 't' is $\frac{5}{2}(17)^{3/2}$ metre. Find the time 't' where 't' is in seconds.

Space for rough work

Chemistry

PART – II

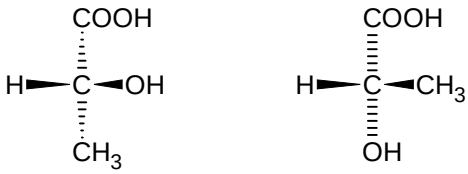
SECTION - A

Straight Objective Type

*This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. Which of the following is the correct nature of a solution obtained by mixing 100 ml of 0.10 M HA and 100 ml of 0.1 M NaOH?
   (I) Neutral, if HA is a strong acid
   (II) Basic if HA is a weak acid
   (III) Neutral if HA is a weak acid.
   (A) I only (B) I and II only
   (C) I and III only (D) II and III only

2. The structures shown below,



   are:
   (A) Identical
   (B) different conformations of the same compound
   (C) Enatiomers
   (D) Diastreomers

3. Consider the following statements:
   (i) 4 g if O_2 gas occupies 11.2 litres at 0^0C and 0.25 atm pressure assuming ideal behaviour.
   (ii) Vander Waal's constant 'a' for $NH_3(g)$ is higher than $CO_2(g)$.
   (iii) Average distance travelled by a gas molecule between two successive collisions is directly proportional to the square of its collision diameter.
   (iv) Average kinetic energy of two molecules of an ideal gas is $\frac{3}{2}kT$, where k is Boltzmann constant.
   Identify the correct statements of the above
   (A) I, ii and iii (B) I, ii, iii and iv
   (C) ii, iii and iv (D) I, ii and iv

4. If we wish to prepare 250 ml of 0.1 M NaOH solution, it is advisable to use more than 1.0 g (as required) NaOH because,
   (A) NaOH is hygroscopic in nature (B) NaOH is easy to purify
   (C) NaOH is less soluble in water (D) NaOH is a strong base

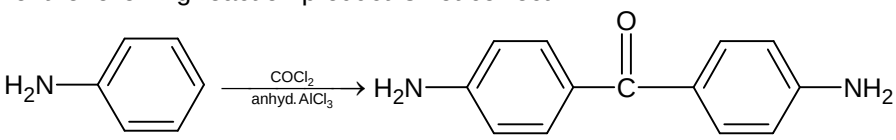
Space for rough work

(Multiple Correct Choice Type)

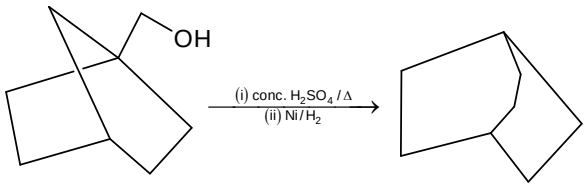
This section contains ***5 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

5. Which of the following reaction product is not correct?

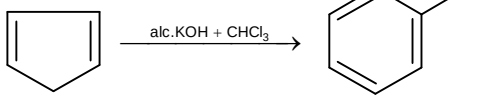
(A)



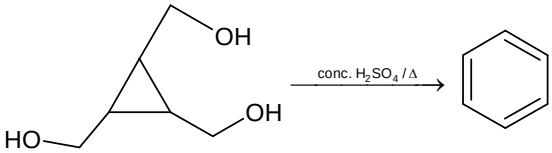
(B)



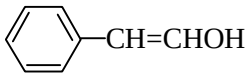
(C)

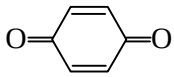


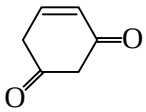
(D)

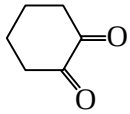


6. Which of the following compounds exhibit tautomerism?

(A) 

(B) 

(C) 

(D) 

Space for rough work

7. In which of the following reaction(s), H_2O_2 can convert the underlined atom into the product with oxidation state = +6.

(A) $\underline{Cr}Cl_3 + H_2O_2 + OH^- \longrightarrow$ (B) $K_2\underline{Cr}_2O_7 + H_2O_2 + H_2SO_4 \longrightarrow$

(C) $\underline{S}O_2 + H_2O_2 + OH^- \longrightarrow$ (D) $H_2O_2 + Na_3\underline{As}O_3 + OH^- \longrightarrow$

8. From the following data for the reaction between A and B:

[A] $[mol^{-1}]$	[B] $[mol^{-1}]$	Initial rate at $(mol^{-1}sec^{-1})$	
		300 K	320 K
2.5×10^{-4}	3.0×10^{-5}	5.0×10^{-4}	2.0×10^{-3}
5.0×10^{-4}	6.0×10^{-5}	4.0×10^{-3}	-
1.0×10^{-3}	6.0×10^{-5}	1.6×10^{-2}	-

Now select the correct statement(s) of the following:

(A) Overall order of this reaction is 3.

(B) Units of rate constant of the above reaction is $mol^{-2}\,l^2\,sec^{-1}$

(C) Rate constant of this reaction at 300 K is $2.66 \times 10^8\,mol^{-2}\,l^2\,sec^{-1}$

(D) Rate constant of this reaction at 320 K is $12.66 \times 10^7\,mol^{-2}\,l^2\,sec^{-1}$

9. The correct statement regarding H_3BO_3 is/are

(A) Monobasic and weak Lewis acid

(B) when mannitol is added to aqueous solution of H_3BO_3 pH of the solution decreases

(C) titration of H_3BO_3 against NaOH is done in presence of mannitol

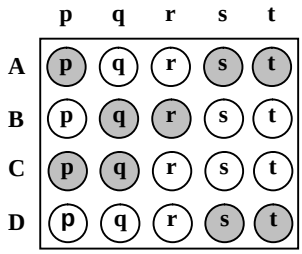
(D) Tribasic and weak Bronsted acid.

Space for rough work

SECTION - B

Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



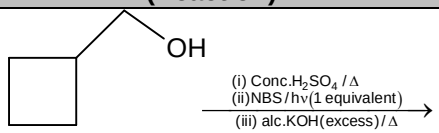
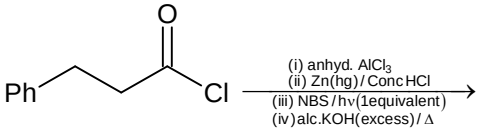
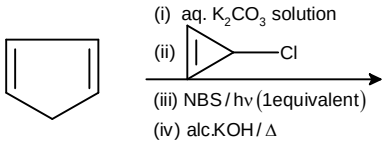
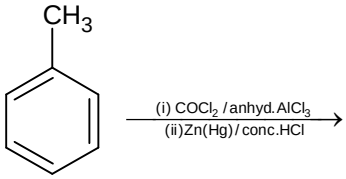
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the species given in Column – I with their corresponding characteristics in Column – II:

Column – I (Species)		Column – II (Characteristics)	
(A)	ICl_2^-	(p)	possesses permanent dipole moment
(B)	$XeOF_4$	(q)	at least two lone pairs on the central atom
(C)	SeF_4	(r)	distorted structure
(D)	$Ba(BrO_3)_2$	(s)	sea-Saw shaped
		(t)	trigonal bipyramidal geometry

Space for rough work

2. Match the reaction in Column – I with the characteristics of corresponding organic product in Column – II:

Column – I (Reaction)		Column – II (Characteristics of organic product)	
(A)		(p)	Aromatic
(B)		(q)	All carbon atoms in the ring are sp^2 hybridised
(C)		(r)	Non-aromatic
(D)		(s)	Bicyclic
		(t)	At least one carbon atom is sp^3 hybridised

Space for rough work

SECTION – C
(Integer value correct Type)

This section contains ***8 questions***. The answer to each question is ***a single digit integer*****,** ranging from 0 to 9 (*both inclusive*).

1. Two moles of helium gas (γ = 5/3) are initially at a temperature of 27^0C and occupy a volume of 20 litre. The gas is first expanded at constant pressure until the volume is doubled. It then undergoes adiabatic change until the temperature returns to its initial value. If total work done by the gas is – xkcal, then what is x?

2. Magnesium carbide on hydrolysis gives a product which on heating with Na followed by treatment with ethyl chloride gives another product (A). (A) on catalytic reduction gives (B). Calculate the number of monochlorinated product (s) of compound (B).

3. 100 ml of tap water containing $Mg(HCO_3)_2$ was titrated with M/50 HCl with methyl orange as indicator. If 30 ml of HCl were required, calculate '**X**', if temporary hardness as parts of $CaCO_3$ per 10^6 parts of water is $\mathbf{X} \times 10^2$.

4. Sodium borohydride on reaction with iodine in the presence of diglyme solution gives a gas (X) which is used as an important rocket fuel. The gas (X) on heating with ammonia forms an adduct which on further heating to high temperature forms another compound (Y).
   Calculate the number of dichloro derivative of compound (Y).

5. How many isomeric dinitrobenzoic acids can exist? $C_6H_3(NO_2)_2CO_2H$

6. What is the number of monochloro substituted derivatives formed from diphenyl methane?

7. In an experiment, 50 ml of 0.1 M solution of a metallic salt reacted exactly with 25 ml of 0.1 M solution of Na_2SO_3. In the reaction SO_3^{2-} is oxidized to SO_4^{2-}. If the original oxidation number of the metal in the salt was 3, what would be the new oxidation number of metal?

8. For the reaction at 25°C
   $A(g) + 2B(g) \rightleftharpoons 2C(g) + D(g)$
   The initial concentration of B is 1.5 times the initial concentration of A. If at equilibrium, the equilibrium concentration of A and D are equal. The equilibrium constant at 25°C is

Space for rough work

Mathematics

PART – III

SECTION - A

Straight Objective Type

*This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. A straight line L with negative slope passes through (8, 2) and cuts the positive axes at P and Q. As L varies, the absolute minimum value of OP + OQ is (O is origin)
   (A) 10 (B) 18
   (C) 16 (D) 12

2. If (α, β) is a point on the circle whose centre lies on the x-axis and which touches the line x + y = 0 at (2, –2), then the greatest value of α is
   (A) $4-\sqrt{2}$ (B) $4+\sqrt{2}$
   (C) $4+2\sqrt{2}$ (D) 6

3. Maximum number of common normal of $y^2 = 4ax$ and $x^2 = 4by$ may be equal to
   (A) 2 (B) 3
   (C) 5 (D) infinite

4. The locus of foot of perpendicular from the centre on any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is
   (A) a circle (B) ellipse
   (C) hyperbola (D) none of these

Space for rough work

(Multiple Correct Choice Type)

This section contains ***5 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

5. The range of values of 'a' such that the angle θ between the pair of tangents drawn from (a, 0) to the circle $x^2 + y^2 = 1$ satisfies $\frac{\pi}{2} < \theta < \pi$, lies in

(A) (1, 2) (B) $\left(1, \sqrt{2}\right)$

(C) $\left(-\sqrt{2}, -1\right)$ (D) $\left(-\sqrt{2}, -1\right) \cup \left(1, \sqrt{2}\right)$

6. In a ΔABC, if $(a + b + c)(b + c - a) = xbc$ then x can be equal to

(A) 1 (B) 2

(C) 3 (D) 4

7. If the equation $\sin^{-1}(x^2 + x + 1) + \cot^{-1}(ax + 1) = \frac{\pi}{2}$ has exactly 2 solutions, then a can not have the integral values

(A) –1 (B) 0

(C) 1 (D) 2

8. If $\cos^4 \theta + \alpha$, $\sin^4 \theta + \alpha$ are the roots of $x^2 + b(2x + 1) = 0$ and $\cos^2 \theta + \beta$, $\sin^2 \theta + \beta$ are the roots of the equation $x^2 + 4x + 2 = 0$, then b is equal to

(A) 1 (B) –1

(C) 2 (D) –2

9. The sum of the coefficients in the expansion of $(2 + 3cx + c^2x^2)^{12}$ vanishes then c equals

(A) –2 (B) 2

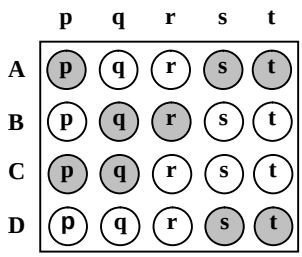
(C) 1 (D) –1

Space for rough work

SECTION - B

Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following Column–I with Column–II

Column – I	Column – II
(A) The function $\frac{\sin x \cos 3x}{\sin 3x \cos x}$ can take the values	(p) 4
(B) $(\sin 12^o)(\sin 48^o)(\sin 54^o)$ is equal to	(q) $\frac{1}{8}$
(C) In an acute angled ΔABC, the least values of $\sum \sec A$	(r) 6
(D) $\left(\sqrt{3}\ \text{cosec}\ 20^o - \sec 20^o\right)$ is equal to	(s) 0
	(t) 2

2. ΔABC has side lengths 18, 24, 30 respectively. Match the following Column–I with Column–II

Column – I	Column – II
(A) The ratio of circum radius to in-radius is	(p) $\sqrt{2}$
(B) The distance between circum-centre and orthocentre is	(q) $\frac{4\sqrt{2}}{3}$
(C) The distance between circum-centre and in-centre	(r) $2\sqrt{2}$
(D) The area of triangle whose vertices are the in-centre, circum-centre and centroid respectively, is	(s) $4\sqrt{2}$
	(t) $\frac{\sqrt{3}}{2}$

Space for rough work

SECTION – C
(Integer value correct Type)

This section contains ***8 questions***. The answer to each question is ***a single digit integer*****,** ranging from 0 to 9 (*both inclusive*).

1. If z_0 lies on the curve $z\bar{z} - (6+i)\bar{z} - (6-i)z + 28 = 0$ then minimum of $|z_0 + 5 - i|$ is equal to _____

2. $|z| = 2$ and the maximum value of $|z-1| + |z-2| + |z-3|$ is M, then $\frac{M}{4}$ is equal to _____

3. Let k be the number of 8-digits number such that the product of any two consecutive digits is prime and $na + \frac{n(n-1)}{2} = k$, where $a, n \in N$. If n is maximum, then $\frac{a}{64}$ is equal to _____

4. 6 boys, 5 girls and 3 teachers are arranged in a line such that boys are in ascending order, girls are in descending order, no two teachers are together. The number of such arrangement is $(220k) \cdot {}^{11}C_5$, then k is equal to _____

5. If 3 positive real numbers a, b, c are in A.P. and abc = 4 then the value of [b], (where [.] denotes greatest integer function) is _____

6. If $\cot(\theta - \alpha)$, $3\cot\theta$, $\cot(\theta + \alpha)$ are in A.P. and θ is not an integral multiple of $\frac{\pi}{2}$, then $\frac{2\sin^2\theta}{\sin^2\alpha}$ is equal to _____

7. If $x, y, z \in (0, 1]$ such that $(\log x)(\log y) = \log(xy)$ and $2(\log x)(\log z) = \log(zx)$, then the value of $2x + 3y + 4z$ is equal to _____

8. $|z^2 - 9| + |z^2| = 41$, then the maximum value of $|z|$ is _____

Space for rough work